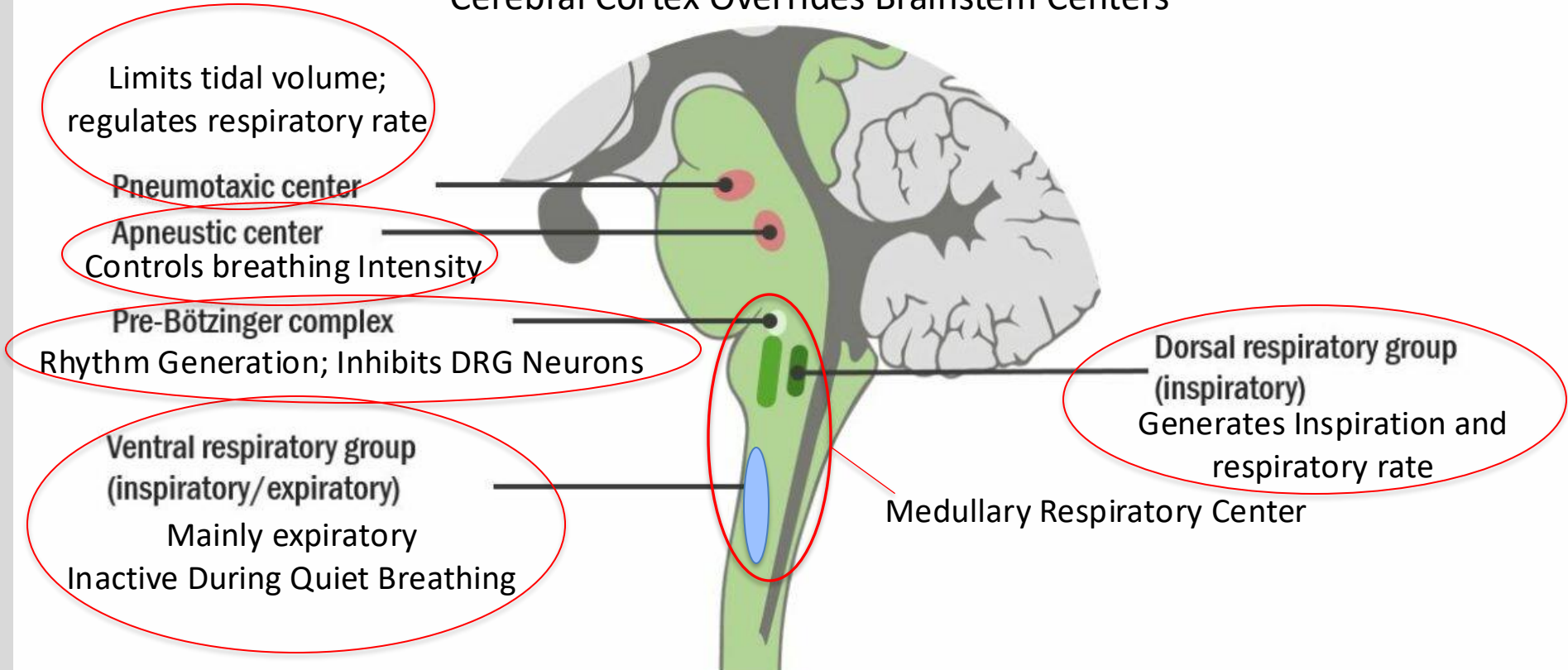


Objectives:

- Name the components of the feedback loop of respiratory control, including phrenic and cranial nerves, and explain how they work together as a unit.
- Name the brainstem centers in control of breathing and outline their basic functions. Compare & contrast focal vs. global hypoxia.
- Explain how the bicarbonate buffer system facilitates the work of the central chemoreceptors of breathing.
- Name the locations where peripheral chemoreceptors can be found and explain their basic characteristics.
- Name other peripheral receptors that influence breathing, what they react to, and how they affect breathing.
- Associate various breathing conditions with their breathing pattern, and potential underlying causes.
- Explain how respiratory physiology adapts to strenuous exercise.
- Explain the effects of high altitude on alveolar and arterial O₂ and how the respiratory system reacts in the short and long term.
- Compare and contrast the causes and effects of hypoxia and hypoxemia on the respiratory system.

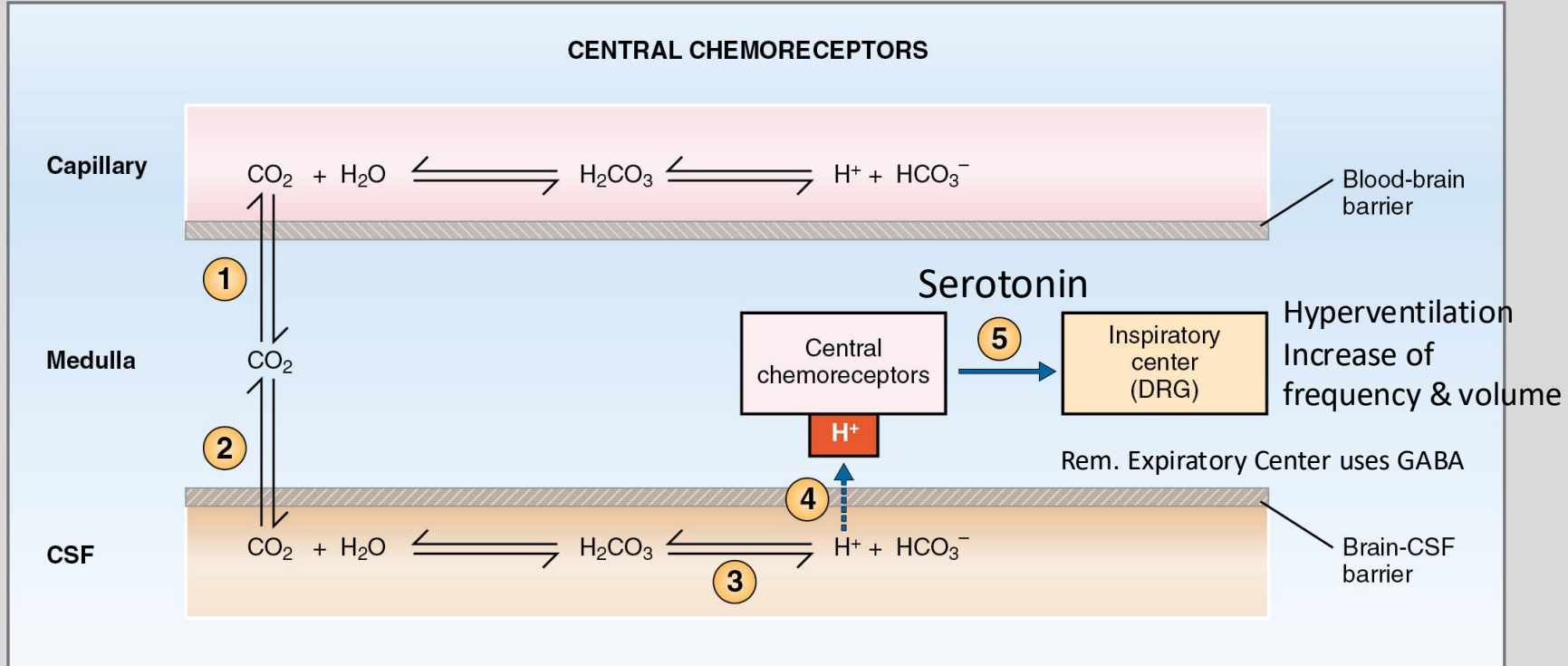
Brain Stem Centers in Control of Breathing

Cerebral Cortex Overrides Brainstem Centers



Medullary Chemoreceptors

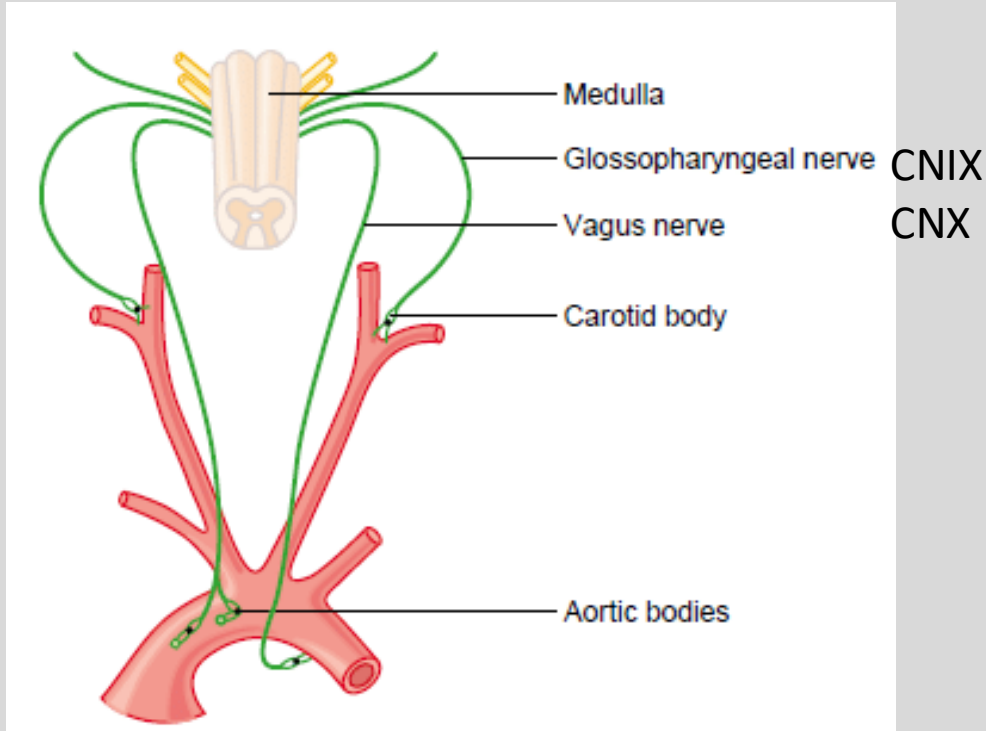
pH of CSF (directly); CO₂ (indirectly)



In sudden infant death syndrome (SIDS), there is a deficit of serotonergic neurons

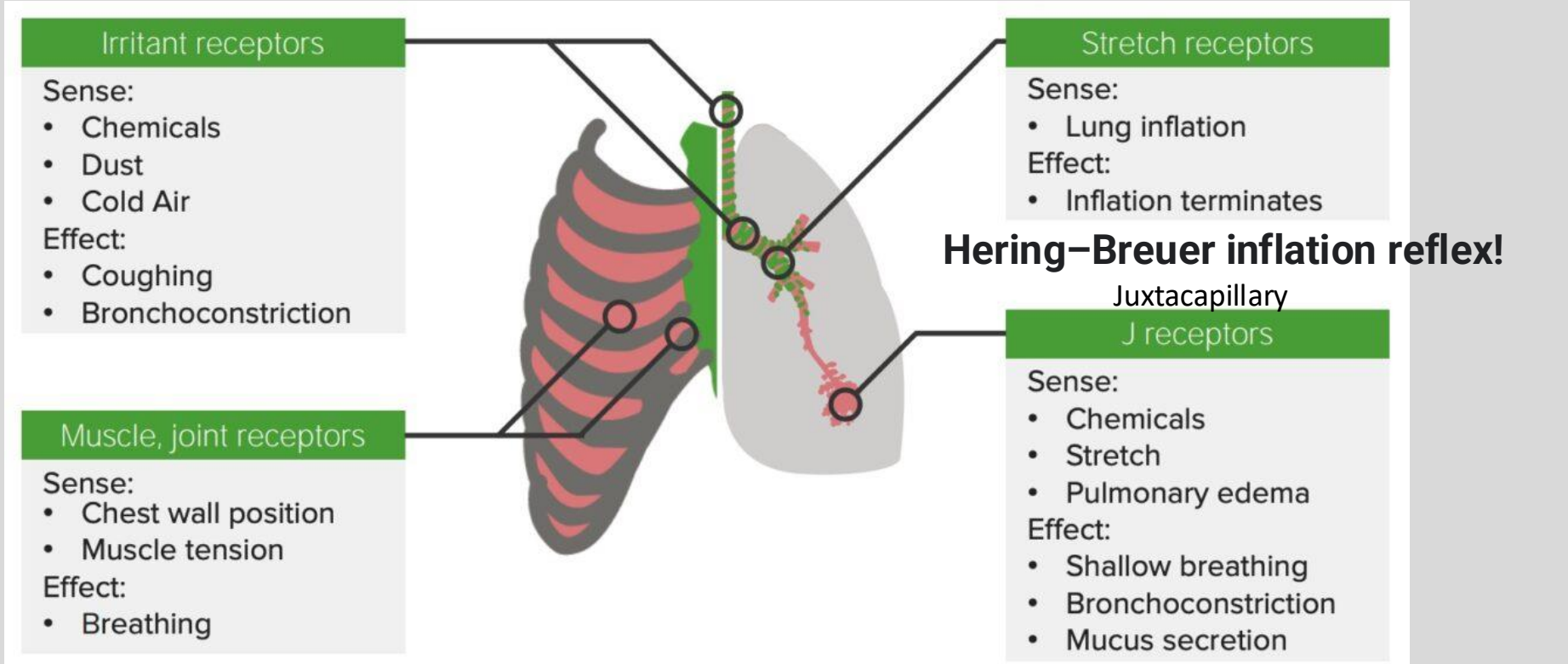
Peripheral Chemoreceptors

O_2 , CO_2 , & H^+



Decreases in PaO_2 (less than 60mmHg)
Increases in $PaCO_2$ (less dominant)
Decreases in pH (carotid bodies only)

Other Peripheral Receptors

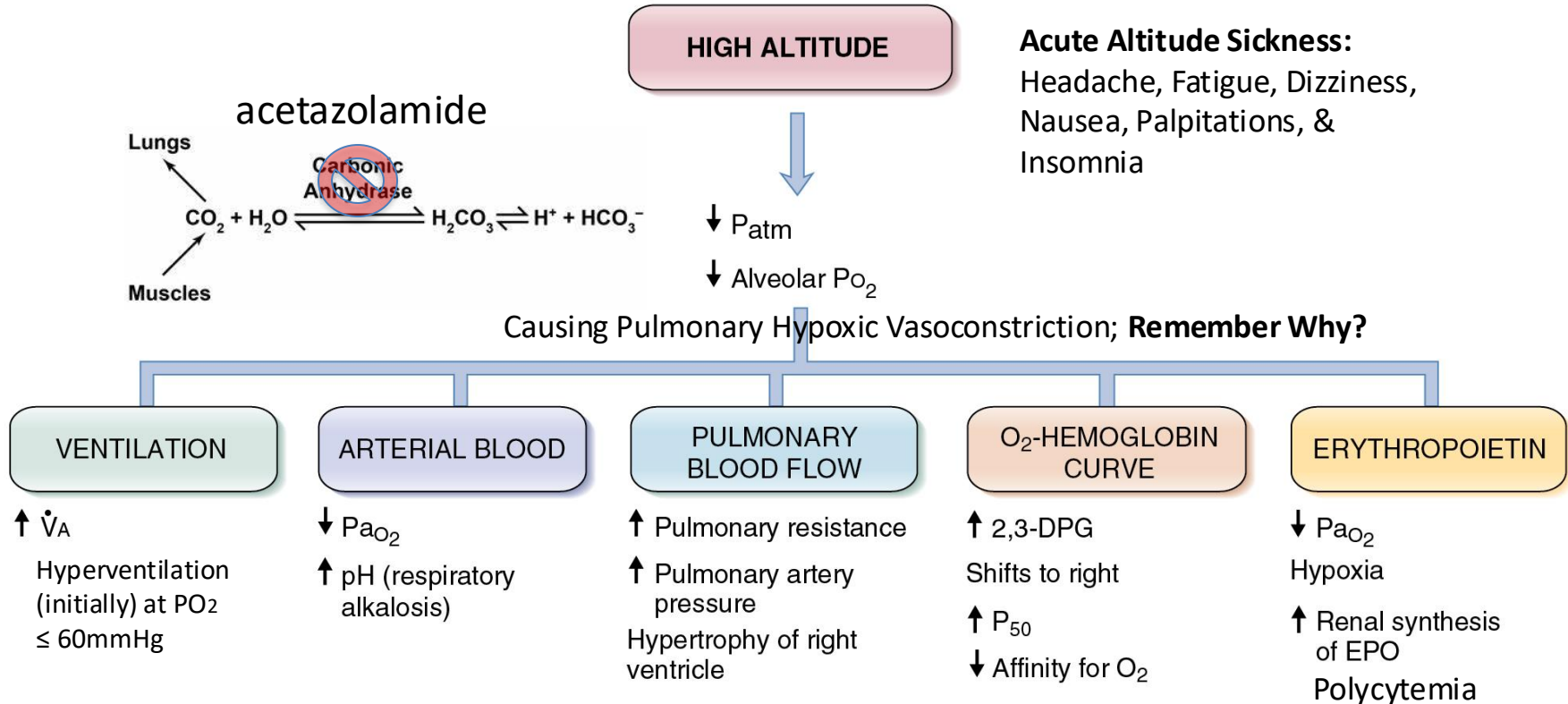


Dyspnea in left-sided heart failure

Adaptions to High Altitude

Mount Everest PO₂ of Inspired Air ~47 mm Hg!

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Hypoxemia – Decrease in PaO₂

Cause	PaO ₂	A – a Gradient	Supplemental O ₂ Helpful?
High altitude (↓ P _b ; ↓ P _i O ₂)	Decreased	Normal	Yes
Hypoventilation (↓ P _a O ₂)	Decreased	Normal	Yes
Diffusion defect (e.g., fibrosis)	Decreased	Increased	Yes
V'/Q' defect	Decreased	Increased	Yes
Right-to-left shunt	Decreased	Increased	Limited
COVID-19	Decreased	Increased	Yes

Hypoxia – Decrease in O₂ delivery to Tissues

Cause	Mechanism	PaO ₂
↓ Cardiac output	↓ Blood flow	—
Hypoxemia	↓ Pa O ₂ ↓ O ₂ saturation of hemoglobin ↓ O ₂ content of blood	↓
Anemia	↓ Hemoglobin concentration ↓ O ₂ content of blood	—
Carbon monoxide poisoning	↓ O ₂ content of blood Left shift of O ₂ -hemoglobin curve	—
Cyanide poisoning	↓ O ₂ utilization by tissues	—

Medical Board-Style Practice Question

During an experimental trial of mechanical ventilation on an anesthetized animal, a researcher temporarily disables the ventilator's safety limits and delivers a sudden, massive volume of air that severely over-inflates the animal's lungs. The researcher observes an immediate, involuntary arrest of the animal's inspiratory effort and a prolongation of its expiratory phase. Which of the following pathways is responsible for this protective respiratory reflex?

- A. J-receptors in the alveolar walls detect the stretch and signal the spinal cord to cause rapid, shallow gasping.
- B. Carotid body chemoreceptors detect the sudden stretch and signal the apneustic center in the pons to promote gasping.
- C. Slowly adapting pulmonary stretch receptors in the airway smooth muscle send inhibitory signals via the vagus nerve (CN X) to the medulla to terminate inspiration.
- D. Irritant receptors in the trachea detect the high pressure and trigger an immediate, prolonged inspiratory gasp to protect the alveoli.
- E. Central chemoreceptors in the fourth ventricle sense the sudden lung inflation and directly block phrenic nerve firing at the spinal level.